



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE**Off-Campus Centre of Nitte (Deemed to be University)****Fourth Semester B.Tech (Minor in Computer Science) Degree Examinations**

May 2025

CS2005-1 – INTRODUCTION TO OBJECT ORIENTED PROGRAMMING

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. What is a class in Java?
A) A blueprint for creating objects
B) A predefined function in Java
C) A data type that stores only primitive values
D) A built-in Java library
2. What keyword is used to create an instance of a class?
A) static
B) void
C) new
D) class
3. Which of the following correctly defines a simple Java class named Car?
A) class Car{
B) public Car{
C) Car class{
D) define class Car{
4. Which of the following correctly defines a method in Java?
A) method myMethod() { }
B) void myMethod() { }
C) define myMethod() { }
D) function myMethod() { }
5. What is the primary purpose of a constructor in Java?
A) To define a class
B) To create an instance of an object
C) To initialize an object when it is created
D) To delete objects from memory
6. What does the this keyword refer to in Java?
A) The current class itself
B) A parent class reference
C) A newly created object
D) A static variable
7. What happens if a constructor is not explicitly defined in a Java class?
A) The class cannot be instantiated
B) Java automatically provides a default constructor
C) The program will throw an error
D) The class will inherit a constructor from Object
8. What is method overloading in Java?
A) Defining multiple methods with the same name but different parameters
B) Calling a method from another method
C) Using multiple return statements in a method
D) Defining a method inside another method
9. How are objects passed as arguments in Java?
A) By value
B) By reference
C) By both value and reference
D) Java does not allow passing objects as arguments
10. What is the return type of a method that returns an object?
A) void
B) Object
C) The class type of the object being returned
D) Any primitive data type

11. What is inheritance in Java?

- A) A process of copying data between objects B) A mechanism where one class acquires properties and behaviors of another class
C) A way to define multiple classes with the same name D) A way to create private variables

12. What is the purpose of the super keyword in Java?

- A) To call a method of the child class B) To call a constructor or method of the parent class
C) To make a class abstract D) To declare a final variable

13. What happens when a class is declared as final in Java?

- A) It can still be extended but not instantiated B) It cannot have any methods
C) It cannot be extended by another class D) It becomes an abstract class automatically

14. Which keyword is used to define an interface in Java?

- A) interface B) class
C) implements D) extends

15. How do you import a package in Java?

- A) include packageName; B) import packageName.*;
C) using packageName; D) import.packageName;

16. Which keyword is used to manually throw an exception in Java?

- A) throws B) throw
C) catch D) try

17. What happens if an exception is not caught in a try block?

- A) The program continues normally B) The exception is ignored
C) The program terminates with a runtime error D) The JVM automatically fixes the error

18. Which package in Java contains classes for file handling?

- A) java.file B) java.util
C) java.nio D) java.io

19. What happens if the specified file does not exist while using FileReader?

- A) A new file is created B) An exception is thrown
C) The program ignores the file and continues execution D) The file is automatically populated with default values

20. Which method is used to find the length of a string in Java?

- A) size() B) getSize()
C) len() D) length()

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

1. a) Discuss the concept of passing objects as arguments in Java. Write a Java program to demonstrate how an object can be passed as an argument to a method, explaining how the method accesses and manipulates the object's data.
b) Explain the uses of the static keyword in Java with an example program.
c) Define a class and an object in Java. Write the general syntax for defining a class and creating an object.
2. a) Define inheritance in Java. Write a program in Java to demonstrate single inheritance, explaining how the base class and derived class interact.

Marks BT* CO* PO*

6	L*2	1	1
6	L3	2	2
4	L1	1	1
6	L2	2	2

- | | | | | | |
|-------|---|---|----|---|---|
| b) | Explain constructor overloading in Java. Write a Java program to demonstrate constructor overloading with multiple constructors and explain how they work. | 6 | L3 | 2 | 1 |
| c) | Discuss the situation of instance variable hiding in Java with an example program. Explain its impact by providing sample output. | 4 | L3 | 1 | 1 |
| 3. a) | Explain how the super keyword is used to call a superclass constructor in multilevel inheritance in Java. Write a Java program demonstrating this concept and explain the flow of constructor execution with sample output. | 6 | L3 | 2 | 2 |
| b) | Outline the uses of final keyword in Java with example code snippets for different scenarios. | 6 | L2 | 2 | 1 |
| c) | Discuss the different access specifiers used in java. | 4 | L2 | 1 | 2 |

Unit – II

- | | | | | | |
|-------|---|---|----|---|---|
| 4. a) | Describe the general form of defining an interface and implementing it in a class. Write a java program demonstrating the process of a class implementing an interface. | 6 | L3 | 3 | 2 |
| b) | Explain the finally block in Java and describe its properties. Write a java program illustrating the usage of finally block in exception handling. | 6 | L3 | 4 | 1 |
| c) | What is a package in Java? Write a java program to define a class inside the package. | 4 | L2 | 3 | 1 |
| 5. a) | What is a class implementing multiple interfaces in Java? Explain the concept with a Java program demonstrating multiple interface implementation. | 6 | L3 | 3 | 1 |
| b) | Explain the general form of exception handling in Java. Discuss the five important keywords used in exception handling. | 6 | L2 | 4 | 2 |
| c) | Describe how packages are imported and used in a program with a suitable example. | 4 | L3 | 3 | 1 |
| 6. a) | Discuss the access protection mechanism in Java packages in detail. | 6 | L2 | 3 | 1 |
| b) | Explain the general form of using the throws keyword in Java. Illustrate the usage of throws keyword with an example program. | 6 | L3 | 4 | 1 |
| c) | What are interface variables in Java? Mention their characteristics and demonstrate their usage with a suitable Java program. | 4 | L2 | 3 | 2 |

Unit – III

- | | | | | | |
|-------|--|---|----|---|---|
| 7. a) | Explain the following string handling methods with an example code snippet
i. length()
ii. split()
iii. toLowerCase()
iv. substring() | 8 | L3 | 5 | 2 |
| b) | Design a Java program to write content into a specified file and subsequently read and display the same content. | 8 | L3 | 5 | 2 |
| 8. a) | Discuss the following file handling methods with an example code snippet
i. delete()
ii. exists()
iii. getAbsolutePath()
iv. list() | 8 | L3 | 5 | 2 |
| b) | Outline a Java program that reads an input string from the keyboard, counts the number of vowels and consonants. Demonstrate its working through sample outputs. | 8 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
